

Polarization-Aware DoA Detection Relying on a Single Rydberg Atomic Receiver

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Abstract—A polarization-aware direction-of-arrival (DoA) detection scheme is conceived that leverages the intrinsic vector sensitivity of a single Rydberg atomic vapor cell to achieve quantum-enhanced angle resolution. Our core idea lies in the fact that the vector nature of an electromagnetic wave is uniquely determined by its orthogonal electric and magnetic field components, both of which can be retrieved by a single Rydberg atomic receiver via electromagnetically induced transparency (EIT)-based spectroscopy. To be specific, in the presence of a static magnetic bias field that defines a stable quantization axis, a pair of sequential EIT measurements is carried out in the same vapor cell. Firstly, the electric-field polarization angle is extracted from the Zeeman-resolved EIT spectrum associated with an electric-dipole transition driven by the radio frequency (RF) field. Within the same experimental cycle, the RF field is then retuned to a magnetic-dipole resonance, producing Zeeman-resolved EIT peaks for decoding the RF magnetic-field orientation. This scheme exhibits a dual yet independent sensitivity on both angles, allowing for precise DoA reconstruction without the need for spatial diversity or phase referencing. Building on this foundation, we derive the quantum Fisher-information matrix (QFIM) and obtain a closed-form quantum Cramér-Rao bound (QCRB) for the joint estimation of polarization and orientation angles. Finally, simulation results spanning various quantum parameters validate the proposed approach and identify optimal operating regimes. With appropriately chosen polarization and magnetic-field geometries, a single vapor cell is expected to

achieve sub-0.1° angle resolution at moderate RF-field driving strengths.

Index Terms—Quantum sensing, Rydberg atomic receivers, direction of arrival (DoA), polarization, quantum Cramér-Rao bound.

I. INTRODUCTION

THE emerging blueprint for sixth-generation (6G) communications calls for centimeter-level positioning accuracy, sub-millisecond latency and pervasive sensing at millimeter-wave and Terahertz frequencies - a performance that pushes today's electromagnetic (EM) receiver technology to its classical limits. Against this backdrop, quantum sensing emerges like dawn-light piercing through the fog of the physical layer designs [1]. In particular, Rydberg atomic receivers exploit the extreme sensitivity of highly-excited alkali atoms to convert incident radio-frequency (RF) fields directly into optical signatures, enabling simultaneous resolution of electric-field amplitude, phase, and polarization information at sub-wavelength scales [2], [3], [4], [5], [6]. This capability of exceeding classical detection theory limits redefines the paradigm of joint multi-dimensional EM parameter extraction.

Owing to their large electric-dipole moments, highly excited Rydberg atoms lend themselves to coupling resonantly with microwave and Terahertz radiation, allowing them to respond to electric fields across a broad frequency spectrum having extreme sensitivity. The high sensitivity of single-cell Rydberg atomic receivers has been experimentally shown to be on the order of $\mu\text{V}/\text{cm}/\sqrt{\text{Hz}}$ using a standard structure [4], [7], [8]. This has also been improved to $55 \text{ nV}/\text{cm}/\sqrt{\text{Hz}}$ based on a superheterodyne structure [9], and the employment of the six-wave mixing technique further boosts its sensitivity to $3.98 \text{ nV}/\text{cm}/\sqrt{\text{Hz}}$ [10]. The sensitivity limit of single-cell Rydberg atomic receivers may even reach $\text{pV}/\text{cm}/\sqrt{\text{Hz}}$ under standard quantum limit conditions [5]. Optical readout of the Rydberg electromagnetically induced transparency (EIT) or Autler-Townes (AT) response allows simultaneous extraction of the incident field amplitude [7], [8], [11], [12], [13], of polarization information [14], [15], [16], [17], as well as of both the magnitude and orientation of a magnetic field [18], [19]. Introducing an RF local oscillator (LO) further eases the direct recovery of the signal phase [9], [20], [21], [22], [23]. We now provide a concise overview of these operating principles and state-of-the-art advances in the field of wireless sensing.

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Electric-Field Amplitude Measurement: The EIT technique is typically employed for microwave metrology. When the RF field applied is resonant with the transition frequency between two Rydberg states, the EIT signal exhibits a characteristic splitting phenomenon known as AT splitting [3]. The strength of the electric field can be directly extracted from the AT splitting interval in the EIT spectrum. Upon introducing an LO, the atomic system acts as a Rydberg atomic mixer, enabling the EIT/AT effects in Rydberg atoms to demodulate a second, co-polarized RF field. The resultant difference frequency, or intermediate frequency, can be optically detected through optically probing the Rydberg atoms [9], [20], [21], [22], [23].

Electric-Field Polarization Measurement: The EIT spectroscopy also supports the polarization measurement of the electric field. The technique presented in [14] employs Rydberg-atom based EIT in rubidium vapor cells to measure the electric-field vector, reaching an angle resolution of 0.5° . In [15], the employment of an additional LO facilitates the construction of a Rydberg-atom based mixer, through which a polarization resolution below 0.5° can be attained. Furthermore, it is established in [17] that for a degenerated $S_{1/2} \rightarrow P_{1/2}$ manifold driven by a linearly polarized RF field, the AT splitting and underlying eigen-energies are invariant when the polarization axis is rotated, which is actually an immediate consequence of the spherical symmetry of the Hamiltonian. In [16], this symmetry can be deliberately broken, e.g., by applying a weak bias field, in which case the AT line-shape becomes polarization-dependent. Such mild symmetry breaking inspires the use of variations in π and $\sigma^\pm$ coupling strengths for RF vector sensing.

Magnetic-Field Measurement: The measurement of magnetic fields, including both their magnitude and orientation, can also be pursued under the same framework. Specifically, EIT resonances are associated with the preparation of a coherent, non-interacting superposition state (a so-called “dark state”), which typically involves several metastable atomic levels, such as the Zeeman sub-levels or hyperfine sub-levels of alkali atoms. In the presence of a pair of probe and coupling optical fields, a twin-photon Raman resonance in a Λ configuration is established, where the magnetic field strength can thus be inferred from Zeeman shifts within the EIT resonance [24]. In [25], magnetic-field-induced splitting of the EIT resonance is observed in a three-level ladder system, modeling the vector response to an RF-modulated magnetic field. Furthermore, the orientation of the static bias-field can be extracted by analyzing the relative strengths of the EIT peaks [18], [19]. More specifically, it is demonstrated in [18] and [19] that in a lin || lin optical EIT vector-magnetometry configuration, the EIT resonance amplitude depends on the particular orientation of a static bias magnetic field. Here, we denote the laser wave vector by $\mathbf{k}_{\text{opt}}$ and the laser polarization by ϵ_L . The resonance amplitude reaches an extremum (typically a maximum) when $\epsilon_L \perp \text{span}\{\mathbf{B}_{\text{bias}}, \mathbf{k}_{\text{opt}}\}$, equivalently, $\epsilon_L \parallel (\mathbf{k}_{\text{opt}} \times \mathbf{B}_{\text{bias}})$. Therefore, by measuring the EIT amplitude while rotating the laser polarization around a fixed $\mathbf{k}_{\text{opt}}$, one can retrieve the orientation of $\mathbf{B}_{\text{bias}}$ (the EIT remains observable without any RF field; only its contrast varies with geometry).

Direction-of-Arrival (DoA) Detection: Several studies have leveraged spatially distributed Rydberg atomic receivers to detect the DoA or angle of arrival (AoA) of incident RF fields [26], [27], [28], [29], [30]. Despite differences in implementation, the core idea in [26], [27], and [28] is fundamentally similar to classical antenna arrays: they utilize multiple vapor cells [26], or arrange multiple probe and coupling beams into laser arrays [27], [29], in order to receive RF signals at various spatial positions. By analyzing the amplitude and/or phase differences of the received signals, these treatises apply direction-finding algorithms for retrieving the DoA of the RF source.

Despite these impressive advances, existing Rydberg-based DoA estimation schemes still face critical limitations. Firstly, their angle resolution inherently scales with the number of sensing elements, but increasing that number is non-trivial. Deploying multiple vapor cells [13], [27] requires phase-coherent detection and meticulous cross-calibration, while each cell’s response is susceptible to local temperature and pressure drifts. Regarding the laser-array based approach implemented in a single vapor cell [29], [31], accurate beam alignment is a requisite and optical cross-talk may significantly erode the performance. Secondly, phase-sensitive techniques rely on an additional LO field; nevertheless, reflections from scatterers may distort the phase-difference-to-DoA mappings. As a remedy, the study in [28] investigates a metal-plate-integrated vapor cell using only a probe-coupling laser pair, achieving an angle resolution of 1.7° without the need for a secondary RF phase reference field.

Inspired by the Rydberg atomic receiver’s intrinsic capability of detecting both polarization and magnetic-field orientation, we propose a DoA detection paradigm using only a probe-coupling laser pair in a single vapor cell. Our approach exploits a fundamental property of plane electromagnetic waves, that is, the wave propagation direction $\mathbf{k}$ is uniquely determined by the orthogonal electric $\mathbf{E}$ and magnetic $\mathbf{B}_{\text{RF}}$ field vectors. By correlating the Zeeman-resolved EIT spectral responses with both the $\mathbf{E}$ -polarization (via electric-dipole transition) and $\mathbf{B}_{\text{RF}}$ -orientation (via magnetic-dipole transition), the DoA can be reconstructed without resorting to spatial diversity or phase referencing. This design harnesses the Rydberg atoms’ native vectorial sensitivity, circumventing the scalability bottlenecks of conventional quantum sensor arrays. The key contributions of this work are summarized as follows.

- We develop an analytical technique for determining the DoA by decoding both the electric-field polarization and Zeeman-resolved magnetic sub-level structure using EIT spectroscopy. A pair of back-to-back EIT measurements is performed in the same single vapor cell and referenced to a single, static bias field that defines the quantization axis for the experimental cycle. Firstly, given an electric-dipole (E1) transition driven by the RF field, the polarization angle θ_{RF} can be extracted from the Zeeman-resolved EIT spectrum. Within the same experimental cycle, the carrier frequency is retuned for driving a magnetic-dipole (M1) transition. The resultant Zeeman-resolved EIT peaks are determined by the spherical-basis

components of the RF magnetic field with respect to the quantization axis; the spectra taken for two bias-field orientations yield the RF magnetic-field orientation θ_B . The proposed scheme exhibits a dual yet independent relationship with this angle pair, facilitating precise DoA reconstruction by jointly addressing these angular dependencies.

- We place the angle-resolution capability of a Rydberg atomic receiver on the same theoretical footing as that of classical antenna arrays. Specifically, commencing from the quantum Fisher-information matrix (QFIM), we derive the quantum Cramér-Rao bound (QCRB) for the joint estimation of the electric-field polarization angle θ_{RF} and the magnetic-field orientation θ_B . Since the DoA detection accuracy of our scheme is ultimately limited by how precisely one can read the EIT spectrum, we identify the squared gradient of the probe-transmission profile as the relevant signal term and define an information-centric signal-to-noise ratio (SNR). This thus leads, under weak-probe conditions, to a compact closed-form QCRB that depends only on the SNR, enabling a direct comparison with the Cramér-Rao bound (CRB) of a conventional uniform linear array (ULA).
- We conduct our quantum-domain simulations using the open-source QuTip quantum-optics toolkit [32], in order to validate the proposed analytical DoA-detection scheme across a number of quantum parameters. We find that, for appropriately chosen polarization and magnetic geometries, a single vapor cell delivers sub-0.1° angle resolution at moderate RF-field driving strengths. However, we also show that either over-driving or under-driving the RF transition erodes performance. When benchmarked against array-based schemes, such as ULA [33] and vector sensor array [34], a single Rydberg atomic receiver outperforms a 16-element array across a low-SNR regime. The QCRB saturates at the SQL floor, beyond which the angle resolution no longer improves upon increasing the SNR, while further enhancement in detection precision can be achieved via tuning intrinsic quantum parameters of the system.

The remainder of this paper is structured as follows. Section II presents the core idea of our proposed scheme. Section III and Section IV demonstrate how the Zeeman-resolved EIT spectrum can be processed for extracting the polarization angle and RF magnetic-field orientation, respectively. Next, Section V quantifies the quantum resolution limits via the QCRB performance. Our simulation results are provided in Section VI, and finally, we conclude in Section VII.

Notation: Throughout this paper, we adopt the following mathematical conventions. In particular, $\text{Tr}(\cdot)$ denotes the trace of a matrix; $(\cdot)^*$ and $(\cdot)^H$ represent the complex conjugate and conjugate transpose (Hermitian transpose) of a scalar, vector, or matrix, respectively; $(\cdot)^{-1}$ denotes the matrix inverse; $\text{vec}(\cdot)$ is the vectorization operator that stacks the columns of a matrix into a column vector; $\|\cdot\|_F$ signifies the Frobenius norm of a matrix. The symbol $\propto$ denotes the proportionality; $\otimes$ and $\oplus$ represents the Kronecker product and the direct sum between two matrices. Regarding

quantum-mechanical notations, Dirac's bra-ket formalism represents quantum states by kets $|\psi\rangle$ and bras $\langle\varphi|$. The expression $\langle\varphi|M|\psi\rangle$ denotes the matrix element of an operator between the initial state $|\psi\rangle$ and the final state $\langle\varphi|$. Furthermore, a reduced matrix element is denoted by double bars, e.g., $\langle m'_J || T || m_J \rangle$, where the double-bar notation indicates independence from the magnetic quantum numbers.

II. CORE IDEA

The core idea in determining the DoA of the incoming RF signal using a single Rydberg atomic receiver relies on the fact that a plane electromagnetic wave is fully specified by a pair of mutually-orthogonal vectors, i.e., the electric-field vector $\mathbf{E}$ and the oscillating RF magnetic-field vector $\mathbf{B}_{\text{RF}}$. By resorting to the electric-field vector $\mathbf{E}$ and the oscillating RF magnetic-field vector $\mathbf{B}_{\text{RF}}$, the propagation vector $\mathbf{k}$ can be uniquely determined, specifically, $\mathbf{k} = \frac{\mathbf{E} \times \mathbf{B}_{\text{RF}}}{\|\mathbf{E} \times \mathbf{B}_{\text{RF}}\|}$.¹ The Rydberg atomic receiver delivers these two angles through two independent spectroscopic signatures recorded in the presence of a known static magnetic bias field $\mathbf{B}_{\text{bias}}$ that defines the quantization axis and delivers Zeeman-resolved magnetic sub-levels.

- *Electric-field polarization angle θ_{RF} :* We determine θ_{RF} via the electric-dipole $L = 1$ (E1) transition $n'S_{1/2} \rightarrow nP_{1/2}$, which is addressed by the RF field in presence of a bias field $\mathbf{B}_{\text{bias}}$. The static magnetic bias field $\mathbf{B}_{\text{bias}}$ lifts the degeneracy of magnetic sub-levels via the Zeeman effect [35], producing a resolvable multi-peak structure in the EIT spectrum. As a result, the excited Rydberg states $n'S_{1/2}$ and $nP_{1/2}$ are split into Zeeman sub-levels $\{|n'S_{1/2}, m'_J\rangle, |nP_{1/2}, m_J\rangle\}$, which can be characterized by the Rabi frequency of the E1-transition Ω_{E1} , scaling with the projection of $\mathbf{E}$ onto the atomic transition dipole $\hat{\mathbf{d}}$

$$\Omega_{\text{E1}}(\theta_{\text{RF}}) \propto \frac{|\hat{\mathbf{d}} \cdot \mathbf{E}(\theta_{\text{RF}})|}{\hbar}, \quad (1)$$

where $\hbar$ is the reduced Planck's constant. By measuring the relative AT splittings associated with linearly (π)-polarized and circularly ($\sigma^\pm$)-polarized components, we directly extract the electric-field polarization angle θ_{RF} .

- *Oscillating RF magnetic-field orientation θ_B :* The orientation θ_B can be determined via the magnetic-dipole $L = 1$ (M1) transition $n'P_{1/2} \rightarrow nP_{3/2}$ driven by an RF electric field in the presence of a bias field $\mathbf{B}_{\text{bias}}$. Similarly, the bias field facilitates both excited Rydberg states to be split into Zeeman sub-levels $\{|n'P_{1/2}, m'_J\rangle, |nP_{3/2}, m_J\rangle\}$, and defines π -polarized and $\sigma^\pm$ -polarized selection rules. Then, the amplitudes of the Zeeman-resolved M1 peaks, which are characterized by the Rabi frequency of the M1-transition $\Omega_{\text{M1}}(\theta_B)$, scale as

$$\Omega_{\text{M1}}(\theta_B) \propto \frac{|\hat{\boldsymbol{\mu}} \cdot \mathbf{B}_{\text{RF}}|}{\hbar}, \quad (2)$$

¹This reconstruction assumes the presence of a dominant radiative plane-wave component, so that the local field is approximately transverse. In strong near-field or comparable-power multipath conditions, the estimate should be interpreted as the local energy-flow direction, which is beyond the single-point and plane-wave-based formulation studied here.

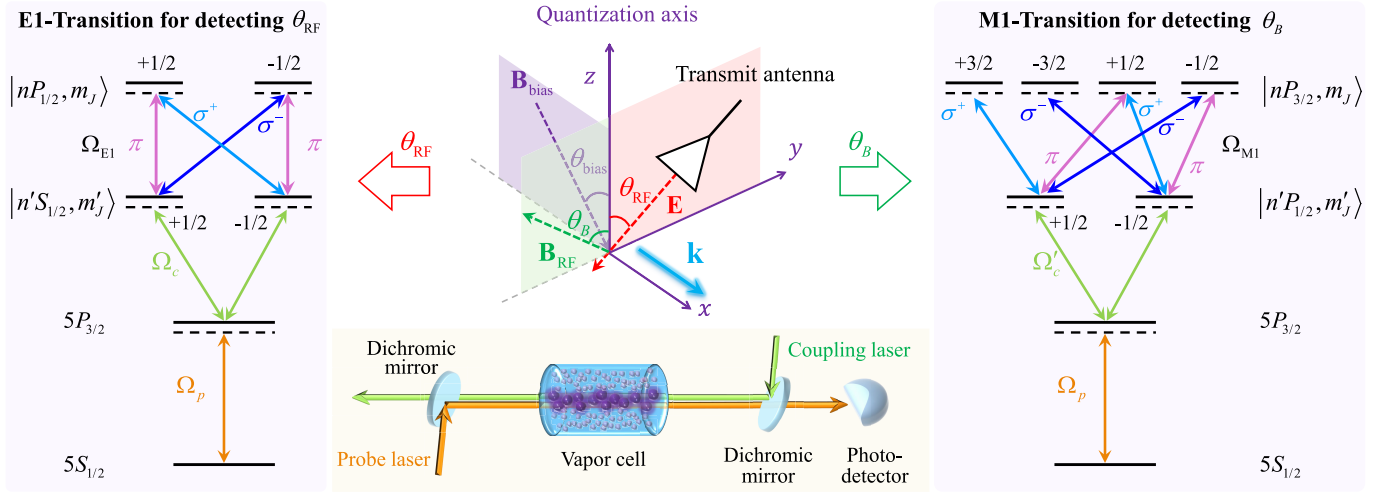


Fig. 1. Illustration of the atomic system in the quantization axis specified by a static magnetic bias field $\mathbf{B}_{\text{bias}}$. The electric-field polarization angle θ_{RF} is inferred from the E1-transition and the RF magnetic-field orientation θ_B is obtained via the M1-transition.

where $\hat{\boldsymbol{\mu}}$ is the magnetic dipole. The oscillating RF magnetic-field orientation θ_B can be identified when θ_{bias} is varied.

Given that we have $\mathbf{E} \perp \mathbf{B}_{\text{RF}}$ for a plane wave in the free space, knowing the angle pair $(\theta_{\text{RF}}, \theta_B)$ is sufficient for reconstructing the full three-dimensional DoA. Once these two vectors are identified, the propagation vector $\mathbf{k}$ follows from $\mathbf{k} = \frac{\mathbf{E} \times \mathbf{B}_{\text{RF}}}{\|\mathbf{E} \times \mathbf{B}_{\text{RF}}\|}$. A single Rydberg atomic receiver can provide both vectors through a pair of independent spectroscopic signatures observed under the same static bias field $\mathbf{B}_{\text{bias}}$. The remainder of this paper derives analytically how θ_{RF} and θ_B imprint themselves on the EIT spectrum of the atomic system considered.

III. DETECTION OF θ_{RF}

In this section, we present a spectroscopic methodology for retrieving the electric-field polarization angle θ_{RF} within a single Rydberg-atom vapor cell. We commence by modeling a four-level atomic system and derive its Hamiltonian. Next, we construct pure Λ -type configurations to resolve the Zeeman sub-levels [19], and derive the analytical expression for an E1-transition's Rabi frequency. Then, based on this, the polarization angle θ_{RF} can be extracted from the Zeeman-resolved peaks.

A. Four-Level Atomic System and Hamiltonian

We consider a four-level atomic system formulated as

$$5S_{1/2} \xrightarrow{\Omega_p} 5P_{3/2} \xrightarrow{\Omega_c} n'S_{1/2} \xrightarrow{\Omega_{\text{E1}}} nP_{1/2}, \quad (3)$$

where a probe laser (Rabi frequency Ω_p) couples the ground state $5S_{1/2}$ to the intermediate excited state $5P_{3/2}$, and a strong coupling laser Ω_c drives the transition from $5P_{3/2}$ up to a Rydberg state $n'S_{1/2}$. Then, the RF field addresses the E1-transition $n'S_{1/2} \rightarrow nP_{1/2}$ whose Rabi frequency is denoted by Ω_{E1} . In this configuration, the probe transmission opens an EIT window whose AT splitting and Zeeman shifts provide direct spectroscopic access to both the amplitude/polarization of the RF field and the orientation of the

oscillating RF magnetic field. As illustrated in Fig. 1, we adopt a right-hand Cartesian laboratory frame in which the azimuth angle 0° lies along the y -axis. In this configuration, the electric-field vector $\mathbf{E}$ lies in the $y-z$ plane and subtends an angle θ_{RF} with respect to the z -axis, while a static magnetic bias field $\mathbf{B}_{\text{bias}}$ defines a known angle θ_{bias} to the same reference. Note that for the measurement of the polarization angle θ_{RF} , a bias field $\mathbf{B}_{\text{bias}}$ is not strictly required, since the probe-coupling beam geometry can itself define the quantization axis. However, we still retain a weak static bias field $\mathbf{B}_{\text{bias}}$ in the $x-z$ plane (yielding an angle θ_{bias} with the positive z -axis) such that the same well-defined quantization axis is used consistently in all electric- and magnetic-field measurements. In our configuration, both the RF wave-vector and the co-propagating probe-coupling beams are aligned along the positive x -axis, which is orthogonal to the plane that contains both $\mathbf{E}$ and $\mathbf{B}_{\text{RF}}$.

The coupling term between the RF field and the atomic dipole moment $\hat{\mathbf{d}}$ can be expressed as the Hamiltonian H_{int} , which is given by

$$H_{\text{int}} = -\hat{\mathbf{d}} \cdot \mathbf{E} = -\langle nP_{1/2}, m_J | \hat{\mathbf{d}} \cdot \mathbf{E} | n'S_{1/2}, m'_J \rangle. \quad (4)$$

Under the rotating wave approximation (RWA) [16], [17], the Hamiltonian of the four-level system considered can be formulated as

$$H_{\text{int}} = \frac{\hbar}{2} \begin{pmatrix} 0 & -\Omega_p & 0 & 0 \\ -\Omega_p^* & -\Delta_p & -\Omega_c & 0 \\ 0 & -\Omega_c^* & -(\Delta_p + \Delta_c) & -\Omega_{\text{RF}} \\ 0 & 0 & -\Omega_{\text{RF}}^* & -(\Delta_p + \Delta_c + \Delta_{\text{RF}}) \end{pmatrix}, \quad (5)$$

where $\Delta_{p,c,\text{RF}}$ represents the detuning associated with the probe laser, coupling laser, and RF field, respectively. Since the RF field generally contains both a component parallel to the quantization axis and components that rotate around that axis, the electric field can be decomposed into the standard spherical basis $(\epsilon_{-1}, \epsilon_0, \epsilon_{+1})$ defined with respect to the quantization axis set by the bias field $\mathbf{B}_{\text{bias}}$. Here ϵ_0 represents the

π -polarized unit vector that drives $\Delta m = 0$ transitions, whereas $\sigma^\pm$ correspond to the $\sigma^\pm$ -circularly-polarized unit vectors that drive $\Delta m = \pm 1$ transitions. When expressing the spherical basis ($\epsilon_{-1}, \epsilon_0, \epsilon_{+1}$) in the right-handed Cartesian coordinate system illustrated in Fig. 1, we obtain $\epsilon_0 = +\mathbf{x}$, $\epsilon_{+1} = \mp \frac{\mathbf{y} \pm j\mathbf{z}}{\sqrt{2}}$, where $\mathbf{x}$, $\mathbf{y}$, and $\mathbf{z}$ represent the unit vectors along the x , y , and z axes, respectively. The RF electric field can therefore be restructured as

$$\mathbf{E} = E_0 [\alpha_{-1}(\theta_{\text{RF}}) \epsilon_{-1} + \alpha_0(\theta_{\text{RF}}) \epsilon_0 + \alpha_{+1}(\theta_{\text{RF}}) \epsilon_{+1}], \quad (6)$$

where E_0 is the electric-field amplitude and the dimensionless coefficients α_q ($q = 0, \pm 1$) obey $\sum_{q=-1}^{+1} |\alpha_q|^2 = 1$. If we focus on the analysis of the $y - z$ plane, i.e., the plane orthogonal to the wave-vector $\mathbf{k} \parallel \mathbf{x}$, the projection of the field onto the spherical basis is purely determined by the angle θ_{RF} between $\mathbf{E}$ and the y -axis

$$\alpha_0(\theta_{\text{RF}}) = \cos \theta_{\text{RF}}, \quad \alpha_{\pm 1}(\theta_{\text{RF}}) = \mp \frac{\sin \theta_{\text{RF}}}{\sqrt{2}}. \quad (7)$$

These definitions explicitly indicate which components of the RF field couple to π ($\Delta m = 0$) or to $\sigma^\pm$ ($\Delta m = \pm 1$) transitions, clarifying the role of θ_{RF} as the polarization angle of interest.

Referring to (4) again, the dipole $\hat{\mathbf{d}}$ can be represented as a linear combination of rank-1 spherical tensor components $\hat{d}_q$ ($q = 0, \pm 1$), which is given by

$$\hat{\mathbf{d}} = \sum_{q=-1}^{+1} \hat{d}_q \epsilon_q. \quad (8)$$

The transition matrix elements in (4) can be further reformulated as

$$\begin{aligned} & - \langle nP_{1/2}, m_J | \hat{\mathbf{d}} \cdot \mathbf{E} | n'S_{1/2}, m'_J \rangle \\ & = -E_0 \sum_{q=-1}^{+1} \alpha_q(\theta_{\text{RF}}) \langle nP_{1/2}, m_J | \hat{d}_q | n'S_{1/2}, m'_J \rangle. \end{aligned} \quad (9)$$

In (9), $\langle nP_{1/2}, m_J | \hat{d}_q | n'S_{1/2}, m'_J \rangle$ can be expanded according to the Wigner-Eckart theorem as follows [36]

$$\begin{aligned} & \langle nP_{1/2}, m_J | \hat{d}_q | n'S_{1/2}, m'_J \rangle \\ & = \langle 1/2, m_J; 1, q | 1/2, m'_J \rangle \langle nP_{1/2} || \hat{d}_q || n'S_{1/2} \rangle \\ & = \sqrt{2} \times (-1)^{1/2-m_J} \begin{pmatrix} 1/2 & 1 & 1/2 \\ -m_J & q & m'_J \end{pmatrix} \langle nP_{1/2} || \hat{d}_q || n'S_{1/2} \rangle, \end{aligned} \quad (10)$$

where $\langle nP_{1/2} || \hat{d}_q || n'S_{1/2} \rangle$ is the reduced dipole matrix element, which is dependent on the magnetic quantum number m_J, m'_J , and the spherical component index q . The term $\langle 1/2, m_J; 1, q | 1/2, m'_J \rangle$ is the Clebsch-Gordan coefficient [37], which can be expanded in the form of a Wigner 3- j symbol formulated as [38]

$$\begin{aligned} & \langle 1/2, m_J; 1, q | 1/2, m'_J \rangle \\ & = \sqrt{2} \times (-1)^{1/2-m_J} \begin{pmatrix} 1/2 & 1 & 1/2 \\ -m_J & q & m'_J \end{pmatrix}. \end{aligned} \quad (11)$$

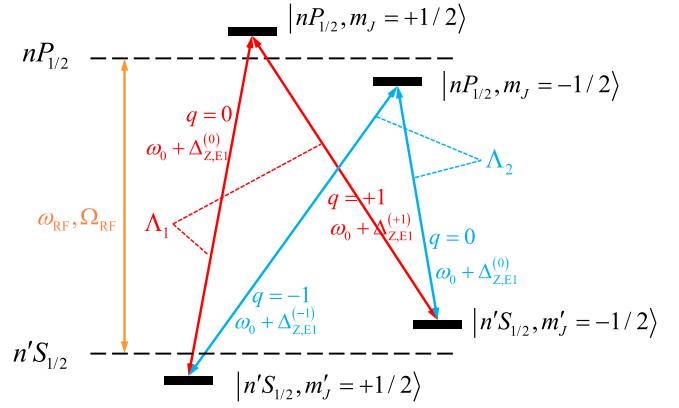


Fig. 2. Illustration of pure Λ systems.

The Clebsch-Gordan coefficients in (11) characterize which transition Δm is allowed and their own contributions. Consequently, (9) can be explicitly written as

$$\begin{aligned} & - \langle nP_{1/2}, m_J | \hat{\mathbf{d}} \cdot \mathbf{E} | n'S_{1/2}, m'_J \rangle \\ & = -E_0 \sum_{q=-1}^{+1} \alpha_q(\theta_{\text{RF}}) \langle nP_{1/2}, m_J | \hat{d}_q | n'S_{1/2}, m'_J \rangle \\ & = -E_0 \sum_{q=-1}^{+1} \alpha_q(\theta_{\text{RF}}) (-1)^{1/2-m_J} \begin{pmatrix} 1/2 & 1 & 1/2 \\ -m_J & q & m'_J \end{pmatrix} \\ & \quad \times \langle nP_{1/2} || \hat{d}_q || n'S_{1/2} \rangle. \end{aligned} \quad (12)$$

B. Pure Λ Systems

In the presence of a weak static magnetic bias field $\mathbf{B}_{\text{bias}}$ characterized by magnitude B_{bias} and orientation θ_{bias} , the excited Rydberg states $n'S_{1/2}$ and $nP_{1/2}$ undergo Zeeman splitting into sub-levels $\{|n'S_{1/2}, m'_J\rangle, |nP_{1/2}, m_J\rangle\}$. When a pair of optical fields couple these sub-levels to a common excited state in a Λ configuration, the atoms are optically pumped into a non-interacting dark superposition, giving rise to Zeeman-resolved EIT resonances whose linewidth and amplitudes depend on the magnetic environment [18], [19]. In this basis, the three spherical components of the RF electric field are mapped onto the relative heights of the Zeeman-resolved EIT peaks. As sketched in Fig. 2, the sub-levels $|n'S_{1/2}, m'_J = +1/2\rangle$ and $|n'S_{1/2}, m'_J = -1/2\rangle$ serve as the Λ ground state, while the common excited state is chosen as $|nP_{1/2}, m_J = \pm 1/2\rangle$. The magnetic selection rules of $\Delta m = 0, \pm 1$ then allow a set of three RF-driven transition paths labeled by the polarization index $q = 0, \pm 1$. A strong $5S_{1/2} \rightarrow 5P_{3/2}$ pump together with an auxiliary repumper optically pumps population into sub-levels $\{|n'S_{1/2}, m'_J\rangle, |nP_{1/2}, m_J\rangle\}$. All other hyperfine levels remain far from being resonant and thus are neglected in our system.

C. E1-Transition Rabi Frequency

The E1-transition Rabi frequency Ω_{E1} can be formulated as

$$\Omega_{\text{E1}}(\theta_{\text{RF}})$$

$$\begin{aligned}
&= \frac{1}{\hbar} \left| \left\langle nP_{1/2}, m_J \right| -\hat{\mathbf{d}} \cdot \mathbf{E} \left| n'S_{1/2}, m'_J \right\rangle \right| \\
&= \frac{E_0}{\hbar} \left| \sum_{q=-1}^{+1} \alpha_q(\theta_{\text{RF}}) (-1)^{1/2-m_J} \begin{pmatrix} 1/2 & 1 & 1/2 \\ -m_J & q & m'_J \end{pmatrix} \right. \\
&\quad \times \left. \left\langle nP_{1/2} \left\| \hat{\mathbf{d}}_q \right\| n'S_{1/2} \right\rangle \right|. \tag{13}
\end{aligned}$$

Furthermore, the RF field does not directly couple the ground state to the probe state; instead, it collects all second-order shifts and broadenings of the intermediate state. Physically, all the transition paths contribute a “self-energy” term Σ_{E1} , which is characterized by

$$\Sigma_{\text{E1}} = \sum_{q=-1}^{+1} \frac{\left| \Omega_{\text{E1}}^{(q)}(\theta_{\text{RF}}) \right|^2}{\Delta_{\text{E1}}^{(q)} + \mathcal{J}\gamma_{\text{RF}}}, \tag{14}$$

The Rabi frequency of each transition path q is given by

$$\begin{aligned}
\Omega_{\text{E1}}^{(q)}(\theta_{\text{RF}}) &= \frac{E_0}{\hbar} \left| \alpha_q(\theta_{\text{RF}}) (-1)^{1/2-m_J} \begin{pmatrix} 1/2 & 1 & 1/2 \\ -m_J & q & m'_J \end{pmatrix} \right. \\
&\quad \times \left. \left\langle nP_{1/2} \left\| \hat{\mathbf{d}}_q \right\| n'S_{1/2} \right\rangle \right|, \tag{15}
\end{aligned}$$

and γ_{RF} is the dephasing rate associated with the E1-transition driven by the RF field. The corresponding detuning for the transition path q is formulated as

$$\Delta_{\text{E1}}^{(q)} = \omega_{\text{RF}} - \left(\omega_{0,\text{E1}} + \Delta_{\text{Z,E1}}^{(q)} \right), \tag{16}$$

where $\omega_{0,\text{E1}}$ denotes the natural transition frequency between the $n'S_{1/2}$ and $nP_{1/2}$ states in the absence of external fields. The Zeeman shift can be expressed as

$$\Delta_{\text{Z,E1}}^{(q)} = \frac{\mu_B B_{\text{bias}}}{\hbar} (g_P m_J - g_S m'_J) \cos \theta_{\text{bias}}, \tag{17}$$

where μ_B is the Bohr magneton, and g_P , g_S represent the Landé factors of the $nP_{1/2}$ and $n'S_{1/2}$ states, respectively.

Note that the RF carrier frequency is typically on the order of Gigahertz separation away from any magnetic-dipole resonance of the Rydberg manifold, and hence the oscillating RF magnetic field acts predominantly as a time-dependent drive, rather than a quasi-static fluctuation. Its only residual influence on the bare energies is a second-order, off-resonant shift, such as the magnetic AC-Zeeman shift [39]. Given the magnitude of the oscillating RF magnetic field $B_{\text{RF}} \lesssim 10^{-7}$ T, the resultant level shift is at most a few Hertz, whose orders of magnitude are much smaller than the kiloHertz-scale static Zeeman shift imposed by the bias field. Therefore, we neglect this AC-Zeeman contribution and treat the RF magnetic component solely as a coupling term that mixes the neighboring Zeeman sublevels.

D. Zeeman-Resolved EIT Spectrum

By accounting for the bias-field-induced Zeeman shift in each transition path, the density matrix element ρ_{21}^{E1} can be explicitly formulated as [22]

$$\rho_{21}^{\text{E1}} = \frac{\mathcal{J}(\Omega_p/2)}{\Delta_p + \mathcal{J}\gamma_{21} - \frac{(\Omega_c/2)^2}{\Delta_c + \mathcal{J}\gamma_{31} - \Sigma_{\text{E1}}}}. \tag{18}$$

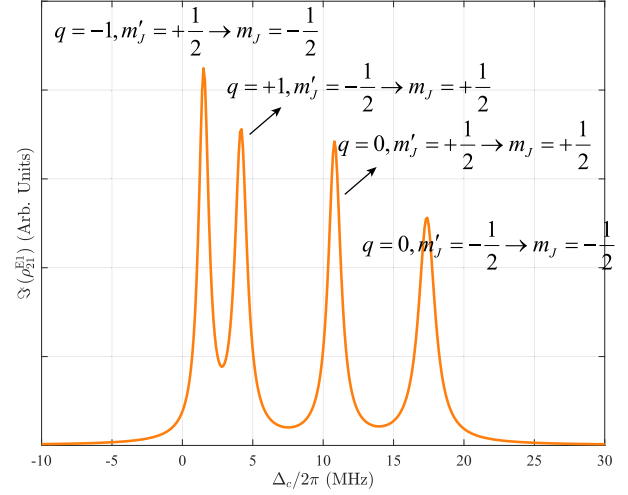


Fig. 3. Zeeman-resolved EIT spectrum at fixed $\theta_{\text{RF}} = 30^\circ$ and $\theta_{\text{bias}} = 0^\circ$.

The above equation encapsulates the components in terms of the RF-dressed self-energy $\Delta_{\text{E1}}^{(q)}$ and the Rabi frequency $\Omega_{\text{E1}}^{(q)}(\theta_{\text{RF}})$ for each transition path q .

To validate the analytic expression given in (18), simulation results are presented in Figs. 3–4 by plotting ρ_{21}^{E1} curves under different configurations. The atomic and optical parameters are chosen to match recent room-temperature vapor-cell experiments [9], [22]: probe and coupling Rabi frequencies of $\Omega_p/2\pi = 0.04$ MHz, $\Omega_c/2\pi = 0.67$ MHz; inverse lifetimes of $\gamma_2/2\pi = 5.2$ MHz, $\gamma_3/2\pi = 3.9$ MHz, and $\gamma_4/2\pi = 0.17$ MHz; as well as zero probe and RF detunings of $\Delta_p = \Delta_{\text{RF}} = 0$. The dipole moment associated with the Rydberg transition is $\hat{\mathbf{d}} = -1443.46e a_0$, with $e = 1.6 \times 10^{19}$ C representing the elementary charge and $a_0 = 5.2 \times 10^{-11}$ m for the Bohr radius. The operation frequency is 6.9 GHz. For a near-resonant E1 transition, the natural transition frequency is treated as almost the frequency of the applied RF field, i.e., $\omega_{0,\text{E1}}/2\pi = 6.9$ GHz. In the presence of detunings, the actual frequency is given by $\omega_{\text{RF}} = \omega_{0,\text{E1}} \pm \tilde{\Delta}$, with $\tilde{\Delta}/2\pi \sim 0.8$ MHz. Given a dephasing rate of $\gamma_{\text{RF}} = 10$ kHz, a static magnetic bias field of $B_{\text{bias}} = 2$ mT, an electric-field amplitude of $E_0 = 1$ V/m, and Landé factors of $g_S = 2.0$ and $g_P = 0.67$, the Zeeman shifts are on the order of MHz, sufficiently high to lift the degeneracy of the four legitimated transitions. Additionally, to extract the polarization (or magnetic field) angle, we rely solely on the relative areas of the Zeeman-resolved AT peaks. As a result, absolute gain calibration is not required. A single reference spectrum acquired at a known angle sets the overall scale, while each subsequent trace is normalized to its simultaneously recorded off-resonant probe transmission.

Specifically, Fig. 3 plots $\Im(\rho_{21}^{\text{E1}})$ versus the coupling detuning Δ_c . As it transpires, four splitting peaks are observed, whose order and amplitudes coincide with the expected Clebsch-Gordan coefficients and spherical-basis projections. Furthermore, in Fig. 4, we retain the same electric-field

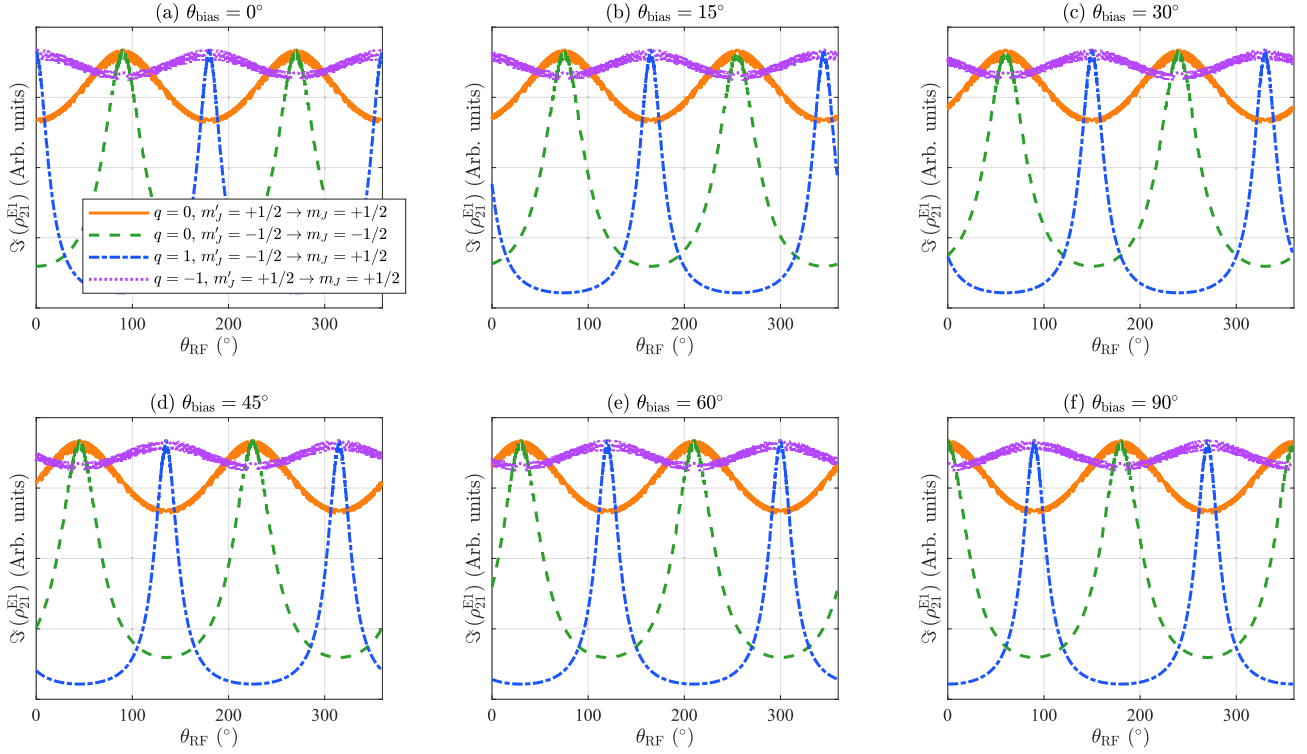


Fig. 4. Zeeman-resolved EIT spectrum with varying θ_{RF} and θ_{bias} .

amplitude of $E_0 = 1$ V/m and sweep the polarization angle θ_{RF} over 360° . Panels (a)–(f) reveal the peak of each Zeeman-resolved transition path for magnetic-field orientations $\theta_{\text{bias}} = 0^\circ, 15^\circ, 30^\circ, 45^\circ, 60^\circ$, and 90° . As it can be observed, under different θ_{bias} , their envelopes follow the expected $|\cos \theta_{\text{RF}}|$ dependence. Regarding the π -polarized case, the two mirror transitions, i.e., $q = 0$, differ in absolute magnitude owing to their different Clebsch-Gordan coefficients for $m_J = \pm 1/2$. For the $\sigma^\pm$ -polarized cases, the maxima occur at $\theta_{\text{RF}} = 90^\circ$ and $\theta_{\text{RF}} = 270^\circ$ when $\theta_{\text{bias}} = 0^\circ$. Tilting the quantization axis away from the optical-beam direction rotates and re-weights the $\sigma^\pm$ versus π admixture: the extrema gradually shift by an amount that scales with $\sin \theta_{\text{bias}}$.

IV. DETECTION OF θ_B

Having demonstrated that the RF-field polarization can be retrieved from the characteristics of the EIT spectrum, we now turn our attention to the reconstruction of the RF magnetic-field $\mathbf{B}_{\text{RF}}$. This section carries out the mirror-image analysis for θ_B 's detection, but on a magnetic-dipole transition basis.

A. Magnetic-Dipole Transition

We replace the atomic system in (3) with the following four-level system

$$5S_{1/2} \xrightarrow{\Omega_p} 5P_{3/2} \xrightarrow{\Omega'_c} n'P_{1/2} \xrightarrow{\Omega_{\text{M1}}} nP_{3/2}. \quad (19)$$

The fine-structured link $n'P_{1/2} \rightarrow nP_{3/2}$ is electric-dipole disabled but magnetic-dipole enabled, and thus the RF field

couples to it only through its magnetic components $\mathbf{B}_{\text{RF}}$. When imposing a bias field $\mathbf{B}_{\text{bias}}$, the excited Rydberg states are split into Zeeman sub-levels $\{|n'P_{1/2}, m'_J\rangle, |nP_{3/2}, m_J\rangle\}$. Pure Λ systems can be constructed, where the two $m'_J = \pm 1/2$ levels of $P_{1/2}$ serve as the Λ ground state and the shared excited state is selected as $|nP_{3/2}, m_J = \pm 1/2, \pm 3/2\rangle$. When $\mathbf{B}_{\text{bias}}$ is rotated away from the quantization z -axis, the definitions of the π ($\Delta m = 0$) and σ ($\Delta m = \pm 1$) couplings are rotated accordingly. This redistributes the population among the Zeeman sub-levels in a manner that encodes the instantaneous orientation of $\mathbf{B}_{\text{bias}}$. In the following, we construct the full Zeeman-resolved Hamiltonian and derive analytical expressions of the M1-transition's Rabi frequency.

B. Magnetic Hamiltonian and M1-Transition Rabi Frequency

The magnetic Hamiltonian H_μ can be formulated as

$$\begin{aligned} H_\mu &= -\hat{\boldsymbol{\mu}} \cdot \mathbf{B}_{\text{RF}} \\ &= -\langle nP_{3/2}, m_J | \hat{\boldsymbol{\mu}} \cdot \mathbf{B}_{\text{RF}} | n'P_{1/2}, m'_J \rangle \\ &= -B_{\text{RF}} \sum_{q=-1}^{+1} \beta_q(\theta_B) \langle nP_{3/2}, m_J | \hat{\mu}_q | n'P_{1/2}, m'_J \rangle \\ &= -B_{\text{RF}} \sum_{q=-1}^{+1} \beta_q(\theta_B) (-1)^{3/2-m_J} \begin{pmatrix} 3/2 & 1 & 1/2 \\ -m_J & q & m'_J \end{pmatrix} \\ &\quad \times \langle nP_{3/2} || \hat{\mu}_q || n'P_{1/2} \rangle, \end{aligned} \quad (20)$$

where the oscillating RF magnetic field $\mathbf{B}_{\text{RF}}$ can be expanded in the same spherical basis $(\epsilon_{-1}, \epsilon_0, \epsilon_{+1})$, yielding

$$\mathbf{B}_{\text{RF}} = B_{\text{RF}} \sum_{q=-1}^{q=+1} \beta_q(\theta_B) \epsilon_q, \quad (21)$$

along with

$$\beta_0(\theta_B) = \cos \theta_B, \quad \beta_{\pm 1}(\theta_B) = \mp \frac{\sin \theta_B}{\sqrt{2}}. \quad (22)$$

To this end, the M1-transition Rabi frequency can be expressed as

$$\begin{aligned} \Omega_{\text{M1}}(\theta_B) &= \frac{B_{\text{RF}}}{\hbar} \left| \sum_{q=-1}^{+1} \beta_q(\theta_B) (-1)^{3/2-m_J} \begin{pmatrix} 3/2 & 1 & 1/2 \\ -m_J & q & m'_J \end{pmatrix} \right. \\ &\quad \times \langle nP_{3/2} \| \hat{\mu}_q \| n'P_{1/2} \rangle \Big|. \end{aligned} \quad (23)$$

Likewise, the self-energy term for the M1-transition, namely Σ_{M1} , which collects the contributions from all allowed transition paths, can be written as

$$\Sigma_{\text{M1}} = \sum_{q=-1}^{+1} \frac{|\Omega_{\text{M1}}^{(q)}(\theta_B)|^2}{\Delta_{\text{M1}}^{(q)} + \mathcal{J}'_{\text{RF}}}. \quad (24)$$

The Rabi frequency $\Omega_{\text{M1}}^{(q)}(\theta_B)$ for the transition path q is in the form of

$$\begin{aligned} \Omega_{\text{M1}}^{(q)}(\theta_B) &= \frac{B_{\text{RF}}}{\hbar} \left| \beta_q(\theta_B) (-1)^{3/2-m_J} \begin{pmatrix} 3/2 & 1 & 1/2 \\ -m_J & q & m'_J \end{pmatrix} \right. \\ &\quad \times \langle nP_{3/2} \| \hat{\mu}_q \| n'P_{1/2} \rangle \Big|, \end{aligned} \quad (25)$$

in which γ'_{RF} denotes the dephasing rate associated with the M1-transition driven by the RF field. The corresponding detuning for the transition path q is given by

$$\Delta_{\text{M1}}^{(q)} = \omega_{\text{RF}} - (\omega_{0,\text{M1}} + \Delta_{\text{Z,M1}}^{(q)}). \quad (26)$$

Consequently, the Zeeman shift becomes

$$\Delta_{\text{Z,M1}}^{(q)} = \frac{\mu_B B_{\text{bias}}}{\hbar} (g_P m_J - g'_P m'_J) \cos \theta_{\text{bias}}, \quad (27)$$

where g_P and g'_P gives the Landé factors for $nP_{3/2}$ and $n'P_{1/2}$, respectively. Therefore, we obtain the Zeeman-resolved EIT spectrum by analyzing the density matrix element of

$$\rho_{21}^{\text{M1}} = \frac{\mathcal{J}(\Omega_P/2)}{\Delta_P + \mathcal{J}'_{21} - \frac{(\Omega'_c/2)^2}{\Delta'_c + \mathcal{J}'_{31} - \Sigma_{\text{M1}}}}, \quad (28)$$

where we use a prime superscript ($'$) to distinguish these variables from those appearing in (18) of the E1-transition.

C. Reconstruction of θ_B From Bias-Field Scanning

Having derived the M1-transition Rabi frequency and the corresponding Zeeman-resolved EIT spectrum, we now discuss how a controlled rotation of the static bias field enables a full reconstruction of θ_B . The procedure consists of two deterministic steps.

- 1) Let $\hat{n}_i$ ($i = 1, 2, \dots$) be a set of bias-field orientations produced by a three-axis Helmholtz coil. For each orientation $\hat{z}_i$, the spherical basis $\{\epsilon_q^{(i)}\}_{q=0,\pm 1}$ is obtained by a passive rotation of the standard (x, y, z) basis into $\hat{n}_i$. The M1-transition coupling associated with the $\Delta m = q$ Zeeman channel is determined by the spherical-basis component of the RF magnetic field with respect to the quantization axis $\hat{n}_i$. Specifically, we define

$$B_{\text{RF},q}^{(i)} = |\mathbf{B}_{\text{RF}} \cdot \epsilon_q^{(i)}|, \quad q \in \{0, \pm 1\}, \quad (29)$$

which corresponds to the π ($q = 0$) and $\sigma^\pm$ ($q = \pm 1$) components, respectively.

- 2) Selecting two mutually orthogonal bias axes, e.g., $\hat{n}_1 \parallel z$ and $\hat{n}_2 \parallel y$, furnishes an invertible system

$$\begin{bmatrix} \epsilon_{+1}^{(1),T} \\ \epsilon_0^{(1),T} \\ \epsilon_{-1}^{(1),T} \\ \epsilon_{+1}^{(2),T} \\ \epsilon_0^{(2),T} \\ \epsilon_{-1}^{(2),T} \end{bmatrix} \mathbf{B}_{\text{RF}} = \begin{bmatrix} b_{+1}^{(1)} B_{\text{RF},+1}^{(1)} \\ b_0^{(1)} B_{\text{RF},0}^{(1)} \\ b_{-1}^{(1)} B_{\text{RF},-1}^{(1)} \\ b_{+1}^{(2)} B_{\text{RF},+1}^{(2)} \\ b_0^{(2)} B_{\text{RF},0}^{(2)} \\ b_{-1}^{(2)} B_{\text{RF},-1}^{(2)} \end{bmatrix}, \quad (30)$$

where $b_q^{(i)}$ encodes the $0/\pi$ phase (sign) of the complex spherical component $\tilde{B}_{\text{RF},q}^{(i)} = \mathbf{B}_{\text{RF}} \cdot \epsilon_q^{(i)}$. In practice, $b_q^{(i)}$ can be obtained by a phase-sensitive (I/Q) readout referenced to the RF source. Then, the oscillating RF magnetic field $\mathbf{B}_{\text{RF}}$ can be obtained by solving equation (30).

In our single vapor cell platform considered, the electric- and magnetic-field measurements are conducted back-to-back by rapid frequency retuning, with no mechanical changes to the vapor cell, bias field, or beam alignment. More specifically, first, the probe laser is locked to $5 S_{1/2} \rightarrow 5 P_{3/2}$ and the coupling laser to $5 P_{3/2} \rightarrow n' S_{1/2}$, while the RF source is tuned to the on-resonance E1-transition $n' S_{1/2} \rightarrow n P_{1/2}$. The resultant Zeeman-resolved EIT spectrum directly yields the polarization angle θ_{RF} . Subsequently, within the same interleaved measurement sequence, an acousto-optic modulator (AOM) shifts the coupling laser in order to address the transition $5 P_{3/2} \rightarrow n' P_{1/2}$, and the RF carrier is retuned from $\omega_{0,\text{E1}}$ to $\omega_{0,\text{M1}}$ for driving an M1-allowed transition $n' P_{1/2} \rightarrow n P_{3/2}$. The prompt electronic switching ensures that both measurements share an effectively identical geometry on a timescale much shorter than experimental drifts, enabling a joint reconstruction of $(\theta_{\text{RF}}, \theta_B)$ without moving parts or beam realignment.

V. QUANTUM CRAMÉR-RAO BOUND (QCRB)

The preceding sections established how the joint electric- and magnetic-field information is characterized on the Zeeman-resolved EIT spectrum. We now ask a more fundamental question: What is the ultimate precision with which those field parameters can be inferred, regardless of the particular measurement strategy? In quantum estimation theory, this limit is set by the quantum Cramér-Rao bound, which in turn is determined by the quantum Fisher information (QFI)

or, in the multi-parameter case, the QFIM [40], [41]. This metric is introduced to quantify how far quantum resources can push interferometric sensitivity beyond the SQL [42]. It serves as an absolute benchmark against which any practical readout scheme, such as optical, microwave or hybrid, must be evaluated. Therefore, we derive the QCRB in this section to examine its theoretical limit.

A. Theoretical QCRB

The angle pair $(\theta_{\text{RF}}, \theta_B)$ is extracted from two sequential, statistically independent spectra that are measured within the same experimental cycle. Each parameter of interest, θ_{RF} and θ_B , is embedded in its respective density matrix $\rho(\theta_{\text{RF}})$ and $\rho(\theta_B)$. These two states collaboratively form a direct-sum state that characterizes the overall measurement

$$\rho(\theta_{\text{RF}}, \theta_B) = \rho(\theta_{\text{RF}}) \oplus \rho(\theta_B). \quad (31)$$

Since the two blocks occupy orthogonal subspaces, the symmetric logarithmic derivatives act independently in each block, and the global QFIM is simply the direct sum of the single-parameter Fisher information, which is formulated as

$$\mathbf{F} = \begin{bmatrix} \mathbf{F}_{\theta_{\text{RF}}} & 0 \\ 0 & \mathbf{F}_{\theta_B} \end{bmatrix} \in \mathbb{R}^{2 \times 2}. \quad (32)$$

As established by Theorem 1 in [40, Eq. (4)], the element $\mathbf{F}_{\theta_i}$ is given in (33), as shown at the bottom of the page.

We then arrive at the QCRB, which is formulated as

$$\text{QCRB}(\hat{\theta}_i) \geq \frac{1}{\nu} \mathbf{F}^{-1}, \hat{\theta}_i \in \{\hat{\theta}_{\text{RF}}, \hat{\theta}_B\}, \quad (34)$$

where ν denotes the number of experimental repetitions, representing the number of independent and identically distributed (i.i.d.) measurement outcomes contributing to the statistical average. Since the QCRB yields a lower bound on the variance, its square root can be treated as the achievable angular resolution, given by $\sqrt{\frac{1}{\nu} \mathbf{F}^{-1}}$, with units of radians. This analytic paradigm exhibits a dual but independent dependence on both θ_{RF} and θ_B . Hence the QFIM determines the joint quantum-enhanced precision for a single-cell DoA reconstruction, without the need for mechanical reconfiguration, spatial diversity, or external phase referencing.

B. SNR-Limited QCRB

In practical measurements, the DoA is inferred by analyzing the EIT spectrum, where the results are inevitably corrupted by various noise sources. Inspired by the classical CRB behavior in ULA-based receivers, where the CRB scales inversely with the SNR, we now introduce an SNR definition tailored to the QFIM. Given that the parameter dependence of the steady-state density matrix $\rho(\theta_i)$ carries the information we expect to extract, we therefore define the sensitivity operator $\mathbf{A}_{\theta_i} = \frac{\partial}{\partial \theta_i} \rho(\theta_{\text{RF}}, \theta_B)$, whose signal power is naturally given

by its squared Frobenius norm $\|\mathbf{A}_{\theta_i}\|_F^2 = \text{Tr}[\mathbf{A}_{\theta_i}^H \mathbf{A}_{\theta_i}]$. In this case, the QFIM in (33) can be trimmed to

$$\mathbf{F}_{\theta_i} = 2[\text{vec}(\mathbf{A}_{\theta_i})]^H [\rho^*(\theta_i) \otimes \mathbf{I} + \mathbf{I} \otimes \rho(\theta_i)]^{-1} \times \text{vec}(\mathbf{A}_{\theta_i}). \quad (35)$$

When the density matrix $\rho(\theta_i)$ exhibits slight variations within the given range of θ_i , or the atomic system is coupled by the weak probe laser, the following approximation holds

$$\rho^*(\theta_i) \otimes \mathbf{I} + \mathbf{I} \otimes \rho(\theta_i) \approx 2\mathbf{I}_{16}. \quad (36)$$

The proof of (36) is given in Appendix A. Consequently, the QFIM $\mathbf{F}$ can be approximated as

$$\mathbf{F} \approx 4 \begin{bmatrix} \|\mathbf{A}_{\theta_{\text{RF}}}\|_F^2 & 0 \\ 0 & \|\mathbf{A}_{\theta_B}\|_F^2 \end{bmatrix}. \quad (37)$$

Upon denoting the total noise by σ_{total}^2 , we can specify the SNR as

$$\text{SNR}_{\theta_i} = \frac{\|\mathbf{A}_{\theta_i}\|_F^2}{\sigma_{\text{total}}^2}, \theta_i \in \{\theta_{\text{RF}}, \theta_B\}. \quad (38)$$

Remark 1: Note that in (38), $\mathbf{A}_{\theta_i} = \frac{\partial}{\partial \theta_i} \rho(\theta_{\text{RF}}, \theta_B)$ inherits the electric-field amplitude E_0 via the Rabi frequency, i.e., $\Omega_{\text{E1}} \propto E_0$. Hence, $\|\mathbf{A}_{\theta_i}\|_F^2$ already scales with the received power encountered by the Rydberg atomic receiver. The QFIM-tailored SNR here is a measurement-agnostic form of “slope-squared over noise”. Under the weak-response approximation and the chain rule from θ_i to Ω_{E1} and E_0 , it reduces to the implementation-level expressions presented in our earlier work [22]. For example, regarding the LO-dressed Rydberg atomic receiver, $\mathbf{A}_{\theta_i} \propto \kappa^2 (\varphi_{\text{RF}}^2 / \hbar^2) P_{\text{Rx}} |h|^2$, such that SNR_{θ_i} coincides with [22] after identifying the optical-electronic gains and adopting the same noise sources.

Then, upon substituting (38) into (37), we obtain

$$\mathbf{F} \approx 4\sigma_{\text{total}}^2 \begin{bmatrix} \text{SNR}_{\theta_{\text{RF}}} & 0 \\ 0 & \text{SNR}_{\theta_B} \end{bmatrix}. \quad (39)$$

We thus arrive at the QCRB as a function of SNR. Additionally, the total noise power σ_{total}^2 includes contributions from both external and internal noise sources [22], [43], [44]. The external noise is primarily from blackbody radiation [43], while the internal noise comprises a diverse range of sources, including quantum projection noise, photon shot noise, photodetector noise, and thermal noise. For exactly modeling these noise components, the motivated readers might like to refer to the discussions in [22], [23], and [44].

VI. SIMULATION RESULTS

A. Configuration

In this section, we present simulation results for characterizing the DoA-estimation precision achieved by a single Rydberg atomic receiver. Most optical and atomic parameters

$$\mathbf{F}_{\theta_i} = 2\text{vec} \left[\frac{\partial}{\partial \theta_i} \rho(\theta_i) \right]^H [\rho^*(\theta_i) \otimes \mathbf{I} + \mathbf{I} \otimes \rho(\theta_i)]^{-1} \text{vec} \left[\frac{\partial}{\partial \theta_i} \rho(\theta_i) \right], \theta_i \in \{\theta_{\text{RF}}, \theta_B\}. \quad (33)$$

are consistent with those used in Figs. 3–4, but we incorporate several deliberate adjustments. All values are based on well-established spectroscopic data to ensure consistency with [9], [17], [19], and [22]. Regarding the E1-transition, the probe and coupling Rabi frequencies are set to $\Omega_p/2\pi = 0.04$ MHz and $\Omega_c/2\pi = 0.67$ MHz, respectively. The inverse lifetimes of states $|2\rangle$, $|3\rangle$, and $|4\rangle$ are given by $\gamma_2/2\pi = 5.2$ MHz, $\gamma_3/2\pi = 3.9$ MHz, and $\gamma_4/2\pi = 0.17$ MHz, so that $\Omega_p/\gamma_{21} \sim 10^{-2}$, and the approximation in Eq. (36) holds with an accuracy better than 1%. We assume zero detuning for the probe laser, i.e., $\Delta_p = 0$. The dipole of the E1-transition is $\hat{d} = -1443.46e a_0$, and for the $|1\rangle \rightarrow |2\rangle$ transition, the dipole moment is $(2.5ea_0)^2$, with $e = 1.6 \times 10^{-19}$ C representing the elementary charge and $a_0 = 5.29 \times 10^{-11}$ m for the Bohr radius. In the E1 step, the RF carrier is set to $\omega_{0,E1}/2\pi \approx 6.9$ GHz (near resonance of the selected E1 Rydberg transition). In the presence of detunings, the actual carrier frequency becomes $\omega_{RF} = \omega_{0,E1} \pm \hat{\Delta}$, where $\hat{\Delta}/2\pi \sim 0.8$ MHz. The static bias field is $B_{\text{bias}} = 2$ mT, with Landé factors $g_S = 2.0$, $g_P = 0.67$, and the electric-field amplitude is $E_0 = 1$ V/m.

As regards to the M1-transition, all optical powers, decay rates and the static bias field are kept identical to the E1-transition measurement so that the two spectra can be acquired sequentially within a single experimental cycle. The carrier frequency is hopped to $\omega_{0,M1}$, which is chosen to lie in the same GHz band for the selected M1-allowed Rydberg transition. The coupling Rabi frequency is retuned to $\Omega'_c/2\pi = 0.8$ MHz, respectively. The Landé factors for $n'P_{1/2}$ and $nP_{3/2}$ are given by $g_S = 0.67$ and $g_P = 1.33$, respectively. Our model assumes a 1 cm vapor cell filled with ground-state atoms at a density of $4.89 \times 10^{16} \text{ m}^{-3}$, held at room temperature 290 K. The two back-to-back E1/M1 spectra are acquired using prompt electronic switching (RF frequency hopping and AOM shifting) on a timescale much shorter than experimental drifts, so that the geometry can be regarded as effectively unchanged during one interleaved measurement sequence. We parameterize the simulations by the local incident RF field at the vapor cell, characterized by E_0 . When desired, E_0 can be mapped to a transmitter power and distance via standard propagation conditions and antenna gain. However, our DoA reconstruction and QCRB analysis are formulated in an input-referred manner and therefore do not rely on a specific link distance. Typical vapor-cell experiments operate at centimeter-to-decimeter antenna-cell separations [26], [31]. From these parameters, one can compute an equivalent effective aperture for the Rydberg receiver, i.e.,

$$A_{\text{eff}} = \frac{4\pi Z_0 N_{\text{atoms}} \hat{d}^2 \omega_{RF} T_2}{\hbar}, \quad (40)$$

whose physical meaning reflects the quantum-mechanical energy exchange efficiency between the atoms and the incident RF field. One can refer to [22, Appendix B] for detailed derivations. All density-matrix evolutions and spectrum calculations were carried out using the QuTiP toolkit [32], including the full Zeeman sub-level structure of the four-level system considered.

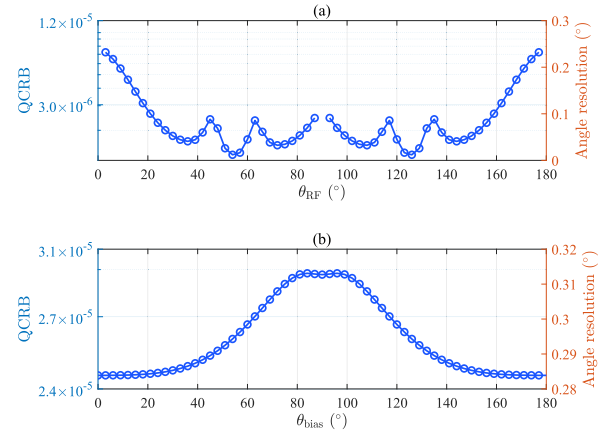


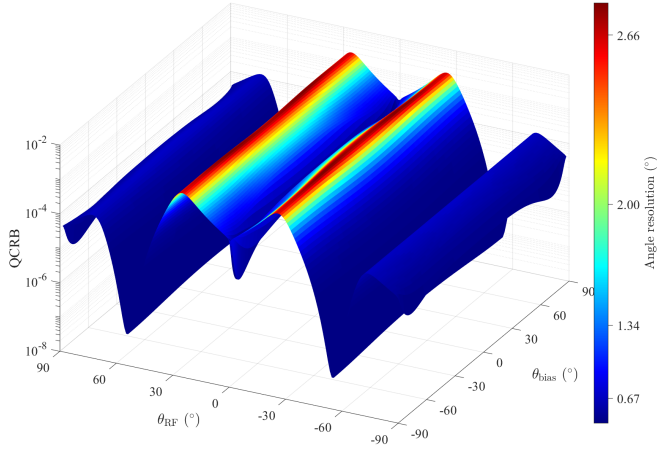
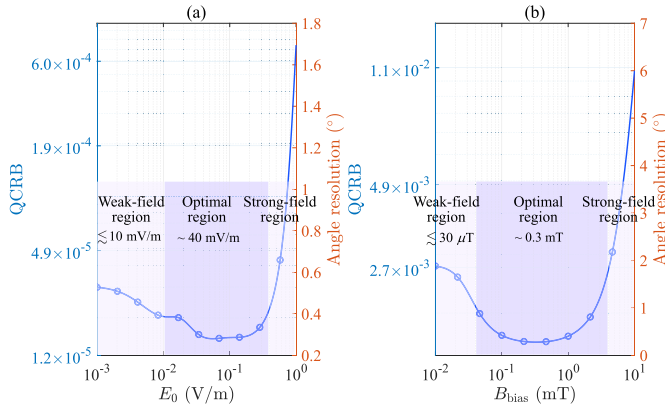
Fig. 5. QCRB versus (a) the polarization angle θ_{RF} when $\theta_{\text{bias}} = 45^\circ$ and (b) the orientation of the bias field θ_{bias} when $\theta_{RF} = 30^\circ$.

B. QCRB Performance

In this subsection, we examine the QCRB performance versus the polarization angle θ_{RF} and the bias-field orientation θ_{bias} . Given the approximation in (37), the QFIM is diagonal and yields two distinct marginal bounds, i.e., $\text{QCRB}(\theta_{RF}; \theta_{\text{bias}}) \geq 1/(\nu \mathbf{F}_{11})$ and $\text{QCRB}(\theta_B; \theta_{\text{bias}}) \geq 1/(\nu \mathbf{F}_{22})$. A dual- y axis is used for visualization: the left y -axis shows the QCRB performance, while the right y -axis gives the corresponding angle resolution in degrees. In Fig. 5(a), we plot $\text{QCRB}_{\theta_{RF}}$ by sweeping θ_{RF} with the bias-field orientation fixed at $\theta_{\text{bias}} = 45^\circ$. More specifically, the QCRB curve is strictly mirror-symmetric about θ_{RF} and truncated at $\theta_{RF} = 0^\circ, 90^\circ, 180^\circ$, because the QCRB explodes to 10^{10} there. The reason for this pronounced divergence lies in a pair of extreme polarization conditions. (i) $\theta_{RF} = 0^\circ, 180^\circ$: the RF field is parallel to the normal of the reference plane and carries only the π component; (ii) $\theta_{RF} = 90^\circ$: the RF field lies inside the reference plane and carries only the $\sigma^\pm$ components. For any case, only a single Zeeman component (π , σ^+ or σ^-) is driven. As a result, at least one of the Zeeman-resolved transitions (i.e., $q = 0, q = \pm 1$) carries zero dipole strength, causing the Fisher information $\mathbf{F} \propto \mathbf{A}_{\theta_i}$ to approach zero, and consequently, the QCRB diverges toward infinity. Once the polarization deviates from these “forbidden” angles, the four transition paths begin to contribute, and the information content recovers rapidly. This gives rise to a series of local minima in the QCRB approximately every $\sim 20^\circ$, each corresponding to a locally optimal angle resolution of $\sim 0.02^\circ$.

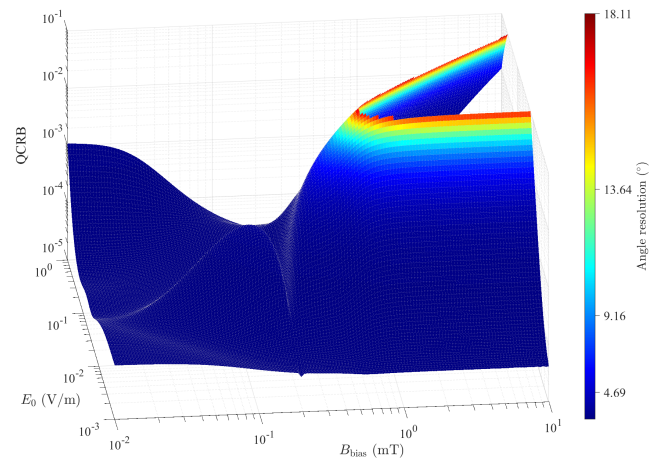
With the polarization angle fixed at $\theta_{RF} = 30^\circ$, the curve in Fig. 5(b) is almost double-peaked. The best angle resolution occurs when $\theta_{\text{bias}} = 0^\circ, 180^\circ$, i.e., when $\mathbf{B}_{\text{bias}}$ is orthogonal to the polarization plane and the quantization axis remains close to the laboratory z -axis. As a result, the transmission gradients of all transition paths are superimposed within the detection plane, and the Fisher information is hence maximized. Rotating the field towards $\theta_{\text{bias}} \approx 90^\circ$ redistributes the polarization weights, compresses the slope, and hence degrades the Fisher information by roughly an order of magnitude.

Specifically, Fig. 6 presents the three-dimensional QCRB surface over the range of $\theta_{RF} \in [-90^\circ, +90^\circ]$ and

Fig. 6. QCRB landscape in the $\theta_{\text{RF}} - \theta_{\text{bias}}$ plane.Fig. 7. QCRB versus (a) the electric-field amplitude E_0 when $B_{\text{bias}} = 0.2$ mT and (b) the static bias field B_{bias} when $E_0 = 0.1$ V/m.

$\theta_{\text{bias}} \in [-90^\circ, +90^\circ]$, visualizing the joint-estimation QCRB performance for the angle pair $(\theta_{\text{RF}}, \theta_B)$. A pair of pronounced ridges, arising at $\theta_{\text{RF}} = 0^\circ, \pm 90^\circ$, mark the Fisher-information zeros already inferred from Fig. 5(a). Furthermore, a narrow valley extends along $\theta_{\text{RF}} \approx \pm 45^\circ$; its floor reaches the global minimum ($\sim 0.02^\circ$) when the magnetic field is orthogonal to the polarization plane (i.e., $\theta_{\text{bias}} \approx 0^\circ, 180^\circ$) and gradually rises by an order of magnitude as the field is rotated towards $\theta_{\text{bias}} \approx 90^\circ$. This landscape thus provides a global view: staying away from the polarization “null” ridges ensures operation with the low-lying valley, where sub- 0.05° quantum-limited angle resolution is achieved.

In Fig. 7(a), the static bias field is fixed at $B_{\text{bias}} = 0.2$ mT, while the electric-field E_0 is swept; in Fig. 7(b), E_0 is fixed at 0.1 V/m, while B_{bias} is varied. Both sweeps reveal the three-regime structure. (i) *Weak-field region*: For $E_0 \lesssim 10$ mV/m or $B_{\text{bias}} \lesssim 30 \mu\text{T}$, the Autler-Townes doublet is obscured by the linewidth, the slope of the probe transmission spectrum with respect to the incident angle approaches zero, and the QFIM collapses. (ii) *Optimal region*: When the dressed-state splitting matches the composite linewidth, the differential phase shift per unit angle is maximized, while the quantum projection noise remains sub-dominant. Under our configurations, the optimum appears near $E_0 \approx 40$ mV/m and $B_{\text{bias}} \approx 0.3$ mT,

Fig. 8. QCRB landscape in the $E_0 - B_{\text{bias}}$ plane.

yielding a quantum-limited angle resolution of approximately 0.25° . (iii) *Strong-field region*: Further increasing either E_0 or B_{bias} separates the Zeeman shifts beyond the EIT linewidth, preventing the accumulation of geometric information and hence degrading the QCRB again.

Fig. 8 visualizes the QCRB surface across $E_0 \in [1 \text{ mV/m}, 1 \text{ V/m}]$ and $B_{\text{bias}} \in [10 \mu\text{T}, 10 \text{ mT}]$. As it transpires, the 3D view exhibits a narrow “valley” of minimum QCRB, where the receiver operates at its quantum optimum. Decreasing either E_0 or B_{bias} leads to under-splitting of the EIT peaks and thus yields poor sensitivity; increasing either E_0 or B_{bias} over-splits the resonance and again degrades the angle resolution. Crucially, this valley spans exactly those specific $(E_0 - B_{\text{bias}})$ combinations that preserve the EIT resonance linewidth, so that one can tune the fields to the optimum without broadening the EIT lineshape and hence without degrading the sensor’s frequency-response. This 3D visualization thus provides a practical guide: stay within the valley to maintain sub- 0.2° angle resolution, while retaining the full spectral agility of the Rydberg atomic receiver.

C. Performance Comparison

In this subsection, we juxtapose the fundamental DoA-detection limits of a Rydberg atomic receiver having a single vapor cell with two well-known RF front-end architectures: (i) a conventional half-wavelength ULA-based scalar array, and (ii) an electromagnetic vector-sensor array (VSA) [34].

- ULA: For an N -element ULA, the CRB on the one-dimensional DoA estimate is given by [33]

$$\text{CRB}_{\text{ULA}} \geq \left(2 \cdot \text{SNR} \cdot k^2 \vartheta^2 \cos^2 \theta \sum_{n=0}^{N-1} n^2 \right)^{-1}, \quad (41)$$

where $k = 2\pi/\lambda$ denotes the wavenumber and $\vartheta = \lambda/2$ is the inter-element spacing. The single-element SNR in (41) can be expressed as $\text{SNR} = P_{\text{Rx}}/\sigma_{\text{AWGN}}^2$, where P_{Rx} is the received power and σ_{AWGN}^2 denotes the power of the additive white-Gaussian-noise (AWGN).

- VSA: The VSA being considered is a co-linear array of the same N physical positions, each position

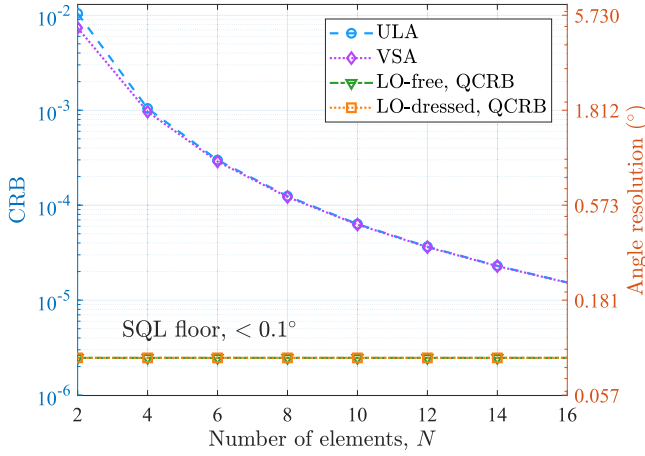


Fig. 9. CRB performance versus the number of elements.

housing a tri-axial electric-dipole triplet that simultaneously records the three orthogonal electric-field components. Under the identical SNR and spacing configurations, the corresponding CRB becomes [34]

$$\text{CRB}_{\text{VSA}} \geq \left[2 \cdot \text{SNR} \cdot \Re \left\{ \left(\frac{\partial \mathbf{a}}{\partial \theta} \right)^H \Xi \left(\frac{\partial \mathbf{a}}{\partial \theta} \right) \right\} \right]^{-1}, \quad (42)$$

where $\Xi = \mathbf{I} - \mathbf{a}(\theta) [\mathbf{a}^H(\theta) \mathbf{a}(\theta)]^{-1} \mathbf{a}^H(\theta)$ and $\mathbf{a}(\theta)$ represents the array response.

As regards to the Rydberg atomic receiver, we consider two architectures: LO-free and LO-dressed (also known as heterodyne [9], [22]) regimes. Their structural difference lies in the presence (LO-dressed) or absence (LO-free) of the local oscillator (LO) drive, as detailed in our earlier work [22]. The DoA is obtained by $\theta^{\text{DoA}} = \text{atan2} \left((\mathbf{E} \times \mathbf{B}_{\text{RF}})_y, (\mathbf{E} \times \mathbf{B}_{\text{RF}})_x \right)$, which is a smooth function. The two-parameter QFIM, obtained from the E1 and M1 transition spectra, is hence convert to a scalar bound via error propagation

$$\text{CRB}(\theta^{\text{DoA}}) \geq \mathbf{J} \mathbf{F}^{-1} \mathbf{J}^T, \quad (43)$$

where $\mathbf{J} = [\partial_{\theta_{\text{RF}}} \theta^{\text{DoA}} \partial_{\theta_{\text{B}}} \theta^{\text{DoA}}]$ is the Jacobi matrix. The CRB is normalized to the same integration time and bandwidth as the array-based case, allowing a direct comparison between the Rydberg QCRB and the array-based CRB under identical conditions.

As it transpires from Fig. 9, the theoretical CRB achieved by the ULA- and VSA-based RF receiver scales as N^{-3} , exhibiting a steep trend. More explicitly, each doubling of the elements reduces the variance by roughly an order of magnitude. For the VSA, the availability of three orthogonal field components per position adds a polarization-derived term to the Fisher information. However, this additional term contributes only a small constant-factor improvement after projection onto the subspace orthogonal to the steering vector. Consequently, the VSA curve is numerically almost indistinguishable from the ULA curve on the plotted scale, and the slope remains determined by the aperture-limited N^{-3} law rather than by the sensor vector dimensionality. In contrast,

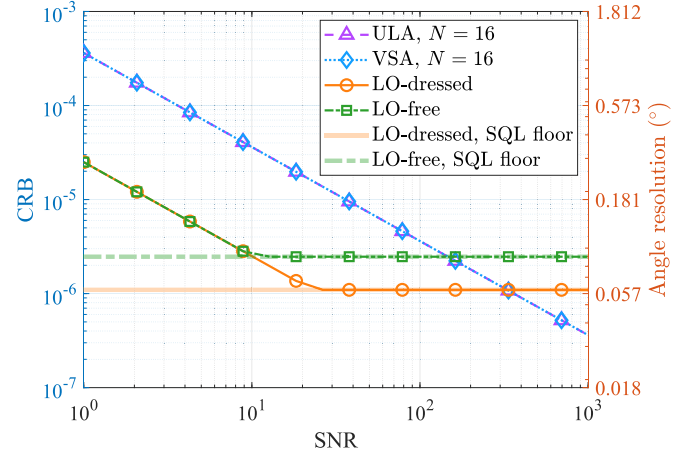


Fig. 10. CRB versus SNR in terms of the Rydberg atomic receiver and the ULA-based RF receiver.

regarding the Rydberg atomic receiver, its fundamental limit is determined by the slope of the EIT response and therefore by the intrinsic optical and atomic parameters, such as the dipole moment $\hat{d}$, coherence time T_2 , dephasing rates γ_{ij} , atom number N_{atoms} , and other related quantities, rather than by its physical size. Fig. 9 demonstrates that a single vapor cell already achieves sub-degree resolution, outperforming both the classical ULA and VSA up to at least $N \leq 16$, while altogether avoiding the hardware scaling and calibration complexity owing to the array aperture growth.

Fig. 10 compares the CRB of array-based receivers and Rydberg atomic receivers as functions of SNR. As expected, the array-based (ULA and VSA) bounds decrease approximately as SNR^{-1} under the ideal AWGN model adopted here. By contrast, the Rydberg receiver exhibits two regimes. At low to moderate SNR (order 10), both LO-free and LO-dressed tracks decrease with SNR and outperform the ULA/VSA because their information is set by the EIT dispersion slope rather than by array aperture. Once the SNR enters the tens-to-tens range, the QCRB tends to a floor and asymptotically approaches the SQL. Beyond this point, improving the SNR cannot translate into finer angle resolution, because the quantum Fisher information is already saturated. Further gains would require increasing the number of atoms, shortening the measurement cycle or enhancing the optical detection efficiency. Additionally, at low SNR, the LO-dressed and LO-free architectures exhibit similar performance. However, owing to its enhanced sensitivity [9], [22], [44], the LO-dressed scheme features an improved SQL, indicating greater SNR potential and thus a tighter QCRB performance.

Fig. 11 plots the CRB performance versus the Tx–Rx distance for different receiver regimes. As observed, the LO-dressed Rydberg atomic receiver achieves the best CRB performance, particularly at larger Tx–Rx distances. This is because the transmission distance primarily affects the received power. Given the same transmission distance, the received signals experience the same fading channel before arriving at the receiver front-end. However, receivers differ in their SNR response to identical received power, with the

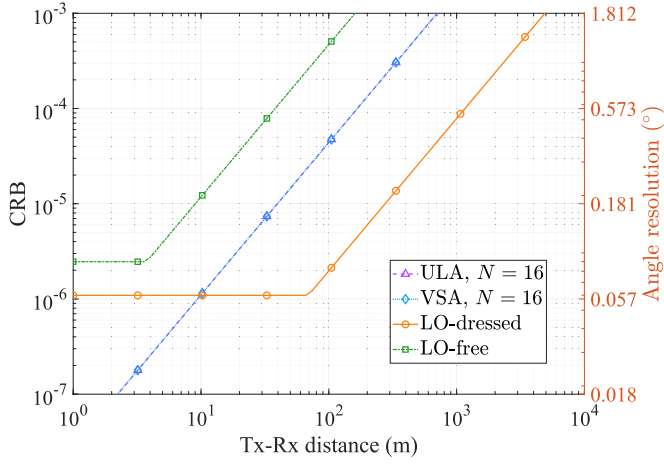


Fig. 11. CRB performance versus the Tx-Rx distance.

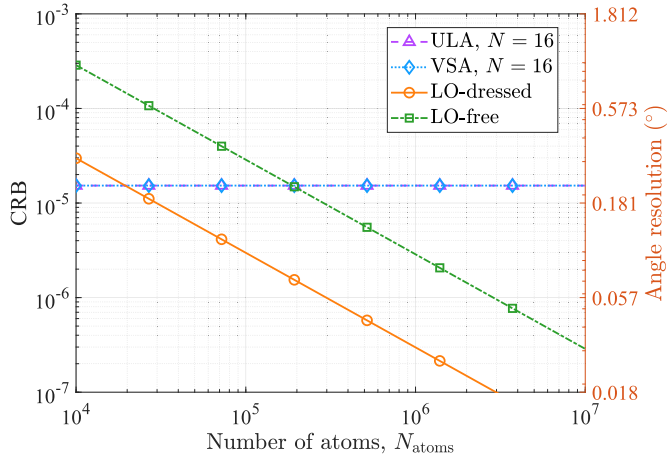


Fig. 12. CRB performance versus the number of atoms.

LO-dressed configuration exhibiting the best SNR response. One may refer to our earlier work for detailed analysis [22]. Accordingly, enhanced SNR leads to improved CRB performance, provided the SQL floor has not yet been reached. Furthermore, both Rydberg atomic receivers clamp to their SQL-determined floors at short ranges, so further SNR increases do not improve the bound. In this saturation regime, the array-based solutions, i.e., ULA and VSA, outperform the Rydberg atomic receivers.

Fig. 12 shows the CRB performance varying with the number of atoms N_{atoms} . In the case of Rydberg atomic receivers, the QCRB decreases as N_{atoms} increases, which is consistent with an SQL imposed by intrinsic optical-atomic parameters, i.e., number of atoms N_{atoms} , rather than by its geometry aperture. Accordingly, once the quantum sensing system nears the SQL floor, further increasing in the SNR may lead to marginal improvements. Enhancements under high-SNR conditions should therefore focus on optimizing intrinsic parameters and optical detection efficiency rather than relying solely on increased RF power.

VII. CONCLUSION

We have conceived a scheme for identifying the DoA of an incident RF signal with sub- 0.1° precision by relying on a single Rydberg atomic vapor cell. The core idea is that a plane electromagnetic wave is fully characterized by its orthogonal electric- and magnetic-field vectors. Based on this, the Zeeman-resolved EIT spectra can be decoded for extracting both the electric-field polarization angle and RF magnetic-field orientation. An analytic expression of this pair of signatures directly reconstructs the incident wave vector. Furthermore, we have demonstrated that the QFIM places the Rydberg counterpart on the same theoretical footing as classical array-based, such as ULA and VSA, RF receivers and specifies a QCRB that depends only on an information-centric SNR. Our simulation results reveal that, in terms of the detection precision quantified by CRB, the Rydberg atomic receiver outperforms 16-element array-based receivers across the low-SNR regime and remains robust to performance erosion even though the RF transition is over-driven.

Future efforts will be devoted to integrated photonic-atomic designs to miniaturize the associated vapor cell architectures, and to extend the dynamic range for high-power RF environments. Experimental validation in multipath-rich scenarios and hybrid quantum-classical fusion architectures also present promising avenues. By bridging quantum sensing and wireless communications, this research opens a transformative path toward atomically precise radio, where the EM field control operates at the fundamental limits of quantum mechanics.

APPENDIX A PROOF OF EQ. (36)

In the weak-probe regime $\Omega_p \ll \gamma_{21}, \Omega_c$, the θ -dependence of the steady-state density matrix is weak, i.e., the probe-induced population redistribution constitutes only a small perturbation. For an algebraically convenient full-rank reference, we write the exact decomposition as follows

$$\rho = \frac{\mathbf{I}}{4} + \varepsilon, \quad (44a)$$

$$\text{Tr}(\varepsilon) = 0, \quad (44b)$$

$$\|\varepsilon\|_F \equiv \varepsilon \ll 1, \quad (44c)$$

and likewise

$$\rho^* = \frac{\mathbf{I}}{4} + \varepsilon^*. \quad (45)$$

Then, we have

$$\rho^* \otimes \mathbf{I} + \mathbf{I} \otimes \rho = \underbrace{\frac{1}{2}\mathbf{I}_{16}}_{\mathbf{X}} + \underbrace{(\varepsilon^* \otimes \mathbf{I} + \mathbf{I} \otimes \varepsilon)}_{\mathbf{Y}}. \quad (46)$$

Since $\|\mathbf{Y}\|_F \leq 4\varepsilon$ and $\|\mathbf{X}^{-1}\|_2 = 2$, we have $\|\mathbf{X}^{-1}\mathbf{Y}\|_F \leq 8 \ll 1$. Inverting $\mathbf{X} + \mathbf{Y}$ via the Neumann series yields that

$$(\mathbf{X} + \mathbf{Y})^{-1} = \mathbf{X}^{-1} \left(\mathbf{I} + (-\mathbf{X}^{-1}\mathbf{Y}) + (-\mathbf{X}^{-1}\mathbf{Y})^2 + \dots \right). \quad (47)$$

Given $\mathbf{X}^{-1} = 2\mathbf{I}_{16}$, the zeroth-order approximation yields

$$[\rho^* \otimes \mathbf{I} + \mathbf{I} \otimes \rho]^{-1} \approx 2\mathbf{I}_{16}, \quad (48)$$

while the first-order correction is

$$[\rho^* \otimes \mathbf{I} + \mathbf{I} \otimes \rho]^{-1} \approx 2\mathbf{I}_{16} - 4\mathbf{Y} + \mathcal{O}(\varepsilon^2). \quad (49)$$

In the QFIM, the diagonal element obeys $\mathbf{F}_{\theta_i} \propto [\text{vec}(\mathbf{A}_{\theta_i})]^H \text{vec}(\mathbf{A}_{\theta_i})$, so the first-order term contributes $\mathcal{O}(\varepsilon)$. For typical parameters $\Omega_p/\gamma_{21} \sim 10^{-2}$, one finds $\varepsilon \lesssim 10^{-2}$ and hence the relative error is below 1%. This thus establishes the validity of Eq. (36).

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